

MemDistill: Diagnosing Why Distilling RL Agent Memory Fails — A Case Study on MEM1 Multi-Hop Retrieval Agents

Anonymous ACL submission

Abstract

We present a *case study* on distilling an RL-trained LLM agent (MEM1) into smaller ReAct-family students, documenting a catastrophic but *silent* SFT distillation failure: students pass tool-call syntax, balanced-XML, and query-similarity checks while retaining <6% of teacher task performance. We introduce a diagnostic protocol—collapse characterization, weight-geometry analysis, and causal intervention—and demonstrate it on MEM1, a 7B RL agent for multi-objective retrieval-augmented QA distilled into Qwen2.5-3B and, to test generality, a cross-family Llama-3.2-1B student. Providing oracle retrieval at inference recovers `em_partial` from 0.021 to 0.498 at $N = 8$ (+47.7pp, $p < 0.0001$); however, a passage-redaction ablation (exp. 016) decomposes this *leakage-inclusive* upper bound into two components, yielding a *partial-mechanism* verdict: retrieval quality contributes ~ 11 pp (23%; +11.0pp [8.8, 13.2] after redacting gold-answer substrings from oracle passages, $p < 0.0001$) as a structural retrieval residual, while passage-level answer-leakage-assisted decoding accounts for the remaining +36.7 pp [+33.5, +40.0] ($\approx 77\%$); capacity accounts for *at most* ~ 12 –14% of the gap (quantified via a 7B full-rank SFT). BM25 retrieval recovers +6–12pp as a deployable mitigation within the teacher’s training document pool. The entire silent-failure signature—format-level indicators near 1.0, content-level metrics (BLEU ≈ 1.15 , ROUGE-L ≈ 0.21) near-floor, and oracle recovery—replicates on Llama-1B, elevating the finding from a case study to a systematic SFT-distillation failure mode.

1 Introduction

SFT distillation of an RL-trained agent can appear to succeed by format-level tool-call metrics—and still fail catastrophically in deployment. We distill MEM1 (Zhou et al.,

2025), a Qwen2.5-7B retrieval-augmented QA agent trained with PPO via `veRL` (Sheng et al., 2024), into a Qwen2.5-3B student using 1,118 teacher trajectories, and we replicate the entire pipeline with a Llama-3.2-1B student (cross-family, $\sim 1/3$ the Qwen-3B student size) to test whether the failure is backbone-specific or systematic (§3.6). On both backbones, format-level tool-call indicators rate the student at 1.00 (tool-call syntax, balanced XML tags, non-empty queries; Qwen query Jaccard = 0.95), while content-level metrics reveal the collapse: BLEU = 1.14/1.16 on sacrebleu’s 0–100 scale, ROUGE-L = 0.21/0.21, and `em_partial` at $N = 8$ retaining only 0.021/0.014 of teacher capability (full per-metric breakdown in Appendix K). Action-sequence accuracy is ≈ 0.606 for both students, but per-sample forensics (Appendix M) reveals this coincidence is structurally degenerate—both students emit the identical action prefix—rather than an independent diagnostic signal. These content warnings are easy to dismiss as cosmetic noise because the student produces teacher-like *structure*—a blind spot mirroring the “syntax-only” tool-call evaluations critiqued by Jhandi et al. (2025) (§4).

It did not succeed on either backbone. The Qwen-3B student achieves `em_partial` = 0.021 at $N = 8$ simultaneous objectives—a 95.4% drop from the teacher’s 0.455—with 83% of outputs truncated by hallucinated search traces. The Llama-1B student reaches `em_partial` = 0.014 at $N = 8$ with 86% truncation—a near-identical collapse despite the change in both family and parameter count. The failure is *silent* at the format/syntax layer on both backbones: the format-level checks all pass while content-level metrics (BLEU, ROUGE-L, `em_partial`) expose the collapse¹. The collapse surfaces only when search

¹Action-sequence accuracy ≈ 0.606 for both students,

calls must return real results, i.e., at pipeline deployment. This is the central danger: an invisible failure—reproducible across backbones—that passes format-level tool-call quality gates while content-level warning signs are overridden by teacher-like structure.

Why do richer training signals not help? At the matched-sample scale available (176 pairs), DPO does not close the gap (+1.0pp, $p = 0.163$). Scaling from 3B to 7B SFT closes only $\sim 12\text{--}14\%$ of the teacher–student gap (§3.2), ruling out capacity as the primary bottleneck.

This paper is deliberately scoped as a *case study* on MEM1-family ReAct agents—a single teacher, a single multi-hop QA domain, and two student backbones—rather than a universal claim about agentic distillation. An earlier attempt to cross-validate the protocol on Kang et al.’s CodeAct-family checkpoints (Kang et al., 2025) (exp_015) was abandoned at the Phase 0 dry-run stage because positional oracle-replay injection presupposes a ReAct-style `<search>/<information>` schema and does not transfer to freeform Python-action protocols; the honest boundary is that our empirical validation is restricted to the ReAct family (§4 Kang paragraph, Appendix N Deviation log).

We introduce a diagnostic protocol that distinguishes *strategy acquisition* from *execution capability*. The key causal test: providing oracle retrieval (referred to throughout as “oracle” for brevity; technically a *teacher-retrieval replay*—we replay the teacher’s actual retrieval output without any answer-key oracle) at inference recovers em.partial from 0.021 to 0.498 at $N = 8$ (+47.7pp, $p < 0.0001$), with truncation dropping from 83% to 0%. This gain is a *leakage-inclusive upper bound*: a passage-redaction ablation (§3.4, Appendix N) yields a *partial-mechanism* decomposition in which retrieval quality alone contributes a *structural retrieval residual* of $\sim 11\text{pp}$ ($\sim 23\%$; +11.0pp after redacting gold-answer substrings from oracle passages, gain measured over the honest_search FlashRAG baseline), while passage-level answer-leakage-assisted decoding contributes the remaining +36.7 pp [+33.5, +40.0] ($\approx 77\%$; Table 2,

but this is a structurally degenerate metric: both backbones emit the identical 3-token action prefix on all 125 test samples (per-sample forensics in Appendix M).

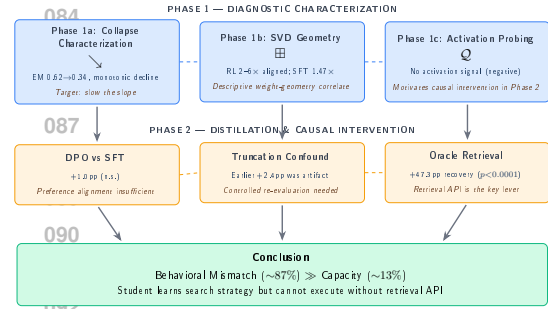


Figure 1: Overview of the MEMDISTILL diagnostic framework. Phase 1 (top) progressively characterizes the distillation failure through collapse analysis, SVD weight geometry, and activation probing. Phase 2 (middle) tests interventions: DPO preference learning, truncation confound correction, and oracle retrieval. The capacity bound (at most $\sim 12\text{--}14\%$) is an independent upper bound; the leakage-inclusive oracle recovery is further decomposed (passage-redaction, exp_016) into retrieval-quality ($\sim 11\text{pp}$, $\sim 23\%$) and passage-level answer-leakage ($\sim 37\text{pp}$, $\sim 77\%$).

Appendix N). Both components are causally meaningful—retrieval is *necessary but not sufficient*—and together establish that the student has learned a viable search strategy but cannot exploit *non-substring-hint* passages at the level its teacher can. Even a simple BM25 retrieval system recovers +6–12pp (within the teacher’s training document pool). The protocol proceeds by elimination—collapse characterization and weight geometry analysis (§2), causal intervention (§3)—and localizes the failure before deployment (Figure 1).

Contributions.

- **A silent failure mode in agentic SFT distillation:** format-level indicators (tool-call syntax 1.00, balanced XML 1.00, query Jaccard 0.95) give high scores while task performance retains $\sim 6\%$ of teacher capability at $N=8$.
- **A diagnostic protocol** (collapse characterization $\rightarrow$ weight geometry $\rightarrow$ causal intervention) that localizes the failure to execution-environment mismatch before pipeline deployment. We propose and demonstrate on MEM1 a three-step elimination scheme (capacity $\rightarrow$ evaluation artefacts $\rightarrow$ causal intervention) for diagnosing silent-failure-shaped gaps.
- **Mechanism decomposition:** oracle (teacher-retrieval replay) +47.7pp ($p < 0.0001$) is

a *leakage-inclusive* upper bound; a passage-redaction ablation (§3.4) decomposes it into retrieval-quality $\sim 11\text{pp}$ (23%) and passage-level answer-leakage $\sim 37\text{pp}$ (77%). Capacity upper-bounds the gap at $\sim 12\text{--}14\%$, acting as an independent upper bound (not a partition); DPO is an informative negative (power > 0.99 at 5pp).

- A **deployable mitigation**: BM25 retrieval recovers $+6\text{--}12\text{pp}$ with no additional training, immediately deployable as a sidecar component within the teacher’s training document pool.

2 Diagnostic Framework

The diagnostic protocol proceeds in two phases. Phase 1 (§2) localises the failure through three progressively narrower analyses; Phase 2 (§3) tests causal interventions distinguishing strategy acquisition from execution capability.

Phase 1: collapse $\rightarrow$ geometry $\rightarrow$ activations.

We evaluate the MEM1 RL policy (Qwen2.5-7B + PPO, trained via veRL (Sheng et al., 2024)) and the Qwen2.5-7B base on a multi-objective RAG QA task from FlashRAG (Jin et al., 2025c); each prompt contains N questions ($N \in \{1, 2, 4, 8, 12, 16\}$) and the agent runs `think` $\rightarrow$ `search` $\rightarrow$ `information` $\rightarrow$ `answer` with per-objective normalized EM (`em_partial`). Three signals together localise the gap. **(1a) Collapse curve.** MEM1’s `em_partial` decreases monotonically from 0.62 at $N = 1$ to 0.341 at $N = 16$ (a 45% drop, $\text{EM} \approx 0.62 - 0.018N$, $R^2 = 0.984$, Cohen’s $d = 0.97$, no inflection point); the base model collapses immediately (0.30 at $N = 1$, $0.075\text{--}0.148$ for $N \geq 2$), so the distillation target is slope reduction rather than cliff recovery (Figure 2). **(1b) Weight geometry (descriptive observational context).** Across all 196 weight matrices of Qwen2.5-7B we measure the alignment ratio $r(k) = \alpha_{\text{obs}}(k) / \alpha_{\text{rand}}(k)$ between fine-tuning deltas $\Delta\mathbf{W} = \mathbf{W}_{\text{ft}} - \mathbf{W}_{\text{base}}$ and the top- k base singular subspace, with α_{rand} calibrated against $n = 100$ Gaussian matrices matched in shape and variance. We observe that RL-base alignment is $2\text{--}6\times$ supra-random (e.g. $6.48\times$ at $k = 64$, $2.78\times$ at $k = 256$), whereas same-architecture full-rank SFT-base reaches only $1.47\times$ at $k = 256$ ($2.92\times$ at $k = 64$); a Wilcoxon signed-rank test gives $p < 1.3 \times 10^{-32}$ at every k and 99% of matrices have $r_{\text{RL}} > r_{\text{SFT}}$

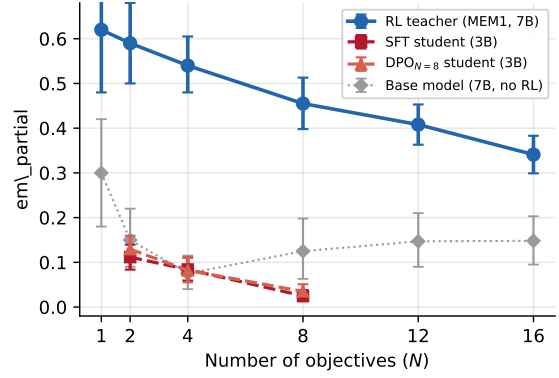


Figure 2: Multi-objective performance collapse. MEM1 degrades monotonically with N ; SFT/DPO students retain $< 8\%$ of teacher capability at $N = 8$; the base model collapses immediately. 95% bootstrap CIs (10,000 resamples).

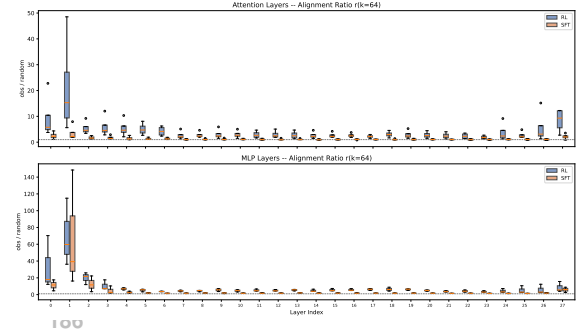


Figure 3: Per-layer SVD alignment ratio $r(k = 256)$ for RL-base and SFT-base deltas across all 196 weight matrices. RL alignment exceeds SFT in 99% of layers (Wilcoxon $p < 1.3 \times 10^{-32}$).

(Figure 3; full per-layer / U-V decomposition in Appendix C). We treat this geometric signal as a descriptive correlate that motivates the causal interventions of §3; the causal role of retrieval execution is established by the oracle ablation, not by this observation. **(1c) Activation probing (negative).** Neither linear probes nor 2-layer MLPs ($n = 300$ prompts, 6 layers, $k \in \{64, 256\}$) recover a per-objective signal from $\Delta\mathbf{h}_l = \mathbf{h}_l^{\text{RL}} - \mathbf{h}_l^{\text{base}}$: linear binary accuracy $0.72 < 0.76$ majority-class, MLP $<$ linear in 47/55 cases ($p = 4.83 \times 10^{-11}$), and 94.5% of readout values fall within a random-projection noise band (Appendix G). We were unable to recover per-objective signal with linear or MLP probes on student activations; we therefore turn to intervention-based analysis in Phase 2 (§3).

3 Distillation Attempt and Failure Analysis

Phase 1 localised a descriptive weight-geometry correlate and a negative activation-probing result. We now test causal interventions to identify where the failure actually lies. We establish DPO and capacity baselines (§3.2), examining the truncation-behavioral-mismatch link (§3.3), and test the causal hypothesis (§3.4).

3.1 Experimental Design

Training pipeline. We train a pooled SFT baseline and three per- N DPO variants on Qwen2.5-3B (LoRA (Hu et al., 2022) rank 32). SFT uses 1,118 partial-credit trajectories (pooled $N \in \{2, 4, 8\}$; total_em ≥ 1) for 3 epochs at $\text{lr} = 2 \times 10^{-4}$. Each DPO variant initialises from the SFT checkpoint, trains on N -specific preference pairs matched to 176 raw pairs at $\beta = 0.05$, $\text{lr} = 2 \times 10^{-5}$, 1 epoch. Full hyperparameters in Appendix A.

Evaluation. Models are evaluated on held-out splits (hash-based, test_frac = 0.25) for $N \in \{2, 4, 8\}$ with max_new_tokens = 4096 (251/131/125 prompts). We report em_partial, em_full, and truncated_rate; significance via paired bootstrap (10,000 resamples); pre-registered practical-significance threshold $\Delta \text{em_partial} \geq +5\text{pp}$ at $N = 8$. The matched-sample constraint subsamples DPO to the $N = 4$ bottleneck of 176 raw pairs—a structural consequence of multi-objective collapse rather than a remediable sampling issue (Appendix H).

Metric: action-sequence accuracy. Let $\mathbf{s} = (s_1, \dots, s_m)$ and $\mathbf{t} = (t_1, \dots, t_n)$ denote the ordered action-tag sequences extracted from student and teacher trajectories respectively, where each tag $\in \{\text{think}, \text{search}, \text{information}, \text{answer}\}$. Action-sequence accuracy is defined as

$$\text{ASA}(\mathbf{s}, \mathbf{t}) = \begin{cases} \frac{1}{n} \sum_{i=1}^{\min(m,n)} \mathbf{1}[s_i = t_i] & n \geq 1, \\ \mathbf{1}[m = 0] & n = 0. \end{cases} \quad (1)$$

The denominator is the *teacher* sequence length, so a student that emits a correct prefix but fewer actions is penalised proportionally. In the *honest_search* setting both

students deterministically emit the 3-token prefix $\langle \text{think}, \text{search}, \text{information} \rangle$ on every sample; the resulting score therefore reduces to $3/n$ and is driven entirely by teacher sequence length, not by student-side variation (per-sample forensics in Appendix M).

3.2 DPO and Capacity Ablations

We also evaluate DPO on matched preference pairs as a stronger-than-SFT baseline; the detailed analysis (Appendix H) confirms DPO yields only +1.0pp ($p = 0.163$, power > 0.99 at 5pp), and does not close the gap at the 176-pair matched-sample scale (power excludes effects $\geq 5\text{pp}$ but remains scale-limited for smaller improvements).

Capacity isolation. A 7B full-rank pooled SFT (same data/inclusion as the 3B LoRA student) consistently outperforms the 3B student (0.171/0.147/0.079 vs. 0.112/0.084/0.025 at $N = 2/4/8$). The oracle gain of +47.7pp at $N = 8$ decomposes on the 3B student oracle axis into retrieval-quality residual ($\approx 11\text{pp}$: oracle_redacted_full – honest_search) plus passage-level leakage ($\approx 37\text{pp}$: oracle_full – oracle_redacted_full). On a separate capacity axis, 7B vs 3B under honest_search contributes $\approx 5\text{--}6\text{pp}$ (absolute $0.079 - 0.025 = 0.054$, or $\approx 12\text{--}14\%$ of the teacher–student gap); this is not additive with the retrieval/leakage decomposition on the oracle axis. Defining the capacity contribution in relative terms as $(\text{em}_{7\text{B}} - \text{em}_{3\text{B}}) / (\text{em}_{\text{teacher}} - \text{em}_{3\text{B}})$, the 7B–3B gap accounts for *at most* 12.3/13.8/12.6% of the teacher–student gap—a stable $\approx 12\text{--}14\%$ upper bound (the 7B uses full-rank fine-tuning, more expressive than rank-32 LoRA, so the training-method advantage is conflated in). This is an upper bound on capacity alone. Crucially, the retrieval-quality bound must now be computed against the *leakage-free* component of oracle recovery, not the full leakage-inclusive +47.7pp: using the passage-redacted oracle (§3.4, exp_016) as the denominator, the non-leakage retrieval-quality upper bound is $\text{em}_{\text{oracle_redacted_full}} - \text{em}_{\text{honest_search}} = 0.131 - 0.021 = 0.110$ at $N = 8$, against which the $\approx 12\text{--}14\%$ capacity bound is no longer a trivial fraction. We therefore interpret capacity as accounting for a non-negligible share of the retrieval-quality-adjusted gap (on the order of compara-

ble magnitude to the retrieval-quality component itself), while passage-level answer-leakage explains the remaining ~ 37 pp of the naive oracle gain.

3.3 Truncation and Behavioral Mismatch

We identified and controlled for a truncation artefact that inflated early DPO signal: without retrieval, both students hallucinate `<search>/<information>` segments that consume $> 80\%$ of the generation budget, causing $\geq 83\%$ output truncation at $N = 8$ (truncation scales monotonically as $30.7\% \rightarrow 55.7\% \rightarrow 83.2\%$ for $N=2/4/8$, with DPO showing the same pattern). A suppression ablation (Appendix J) confirms truncation is the surface manifestation, not the root cause, of behavioral mismatch.

3.4 Causal Verification: Oracle Retrieval

If the failure stems from executing search procedures without a retrieval backend, providing the backend should recover performance. We run two interventions: (i) **BM25 retrieval**, a realistic deployable baseline (our realistic deployable comparison, § *Realistic retrieval: BM25*, Table 3), which replaces the missing backend with a lexical retriever over the teacher’s training document pool; and (ii) **oracle (teacher-retrieval replay)**, an upper-bound probe that intercepts the student’s `<search>` outputs at inference time and injects the passages the RL teacher actually retrieved on the same prompt. We present the oracle result first because it establishes the maximum recoverable headroom; BM25 (Table 3) then quantifies how much of that headroom a realistic system recovers without any answer-adjacent passages. Oracle substantially exceeds BM25 ($+47.7$ pp vs $+12.6$ pp at $N=8$); the redaction ablation below isolates how much of the oracle–BM25 gap is retrieval quality vs passage-level substring leakage.

Table 1 shows oracle retrieval transforms student performance: `em_partial` jumps $0.021 \rightarrow 0.498$ at $N = 8$ ($+47.7$ pp, $p < 0.0001$, 95% CI $[44.6, 50.9]$ pp); truncation drops $83.2\% \rightarrow 0\%$. This provides a *leakage-inclusive upper-bound* on retrieval-only mitigation: the intervention is necessary but not sufficient evidence for a pure retrieval-quality bottleneck, because teacher-replayed passages also contain gold-answer substrings (Appendix D). We note here

	$N=2$	$N=4$	$N=8$
SFT (honest search)	0.112	0.084	0.021
SFT + oracle API	0.530	0.521	0.498
Δ	$+41.8$ pp	$+43.7$ pp	$+47.7$ pp
p (bootstrap)	<0.0001	<0.0001	<0.0001
RL teacher [†]	0.59	0.54	0.455

Table 1: Effect of providing oracle retrieval results at inference time. With real search results injected, the SFT student recovers from near-zero to near-teacher performance. Paired bootstrap, 10,000 resamples. The `honest_search` condition is a real-FlashRAG retrieval baseline (not a `no-search` setting): the student’s `<search>` queries are executed by the production FlashRAG retrieval backend, returning real passages to the student via `<information>` blocks. The `oracle_*` conditions bypass retrieval entirely and inject teacher-replayed passages. The $4.72\times$ latency gap between `honest_search` and `oracle_full` (3666.7s vs 777.4s over 125 prompts) is fully accounted for by FlashRAG retrieval I/O overhead (~ 23 s/prompt; see Appendix D and `exp_016_workdir/no_api_latency_probe.json`).[†]Teacher evaluated on a different prompt set; direct comparison is not valid. Numbers in this table are computed on canonical `phase1d_v2` per- N test partitions ($n_{N=2} = 251$, $n_{N=4} = 131$, $n_{N=8} = 125$); the $N=8$ row reflects `exp_016` re-eval on the identical 125 prompts used by `exp_006` canonical (0.021 vs 0.025 cross-run decoding noise, within ± 0.02 pre-registered tolerance; see Appendix N.6).

that “oracle” is used throughout for brevity; the intervention is technically a *teacher-retrieval replay* (we replay the teacher’s actual retrieval output without any answer-key oracle).

Answer Leakage Audit and Passage-Redaction Ablation. We audit potential answer leakage in teacher information segments. On 50 random $N=8$ evaluation prompts (seed=42, lowercased substring match), 98% of teacher `<information>` blocks contain at least one gold-answer substring; on a derangement-shuffled control (teacher passages from unrelated prompts), the rate is 34% (a 64 pp real-shuffled gap that quantifies retrieval topic specificity). The absolute 98% rate motivates a direct causal test: **passage-redaction** (`exp_016`). On the full `phase1d_v2` $N=8$ test split (125 prompts), we re-evaluate the SFT student with oracle-replay under three redaction conditions—0% (`oracle_full`, sanity-gated to within ± 0.02 of Table 1), $\sim 50\%$ per-passage density (`oracle_redacted_half`, uniform

density halving applied to all 125/125 oracle passages; `redact_info_block_tag` produces 867 substitutions vs 1663 in the full condition, ratio 0.52; see Appendix N and 100% (`oracle_redacted_full`)—replacing gold-answer substrings with a [REDACTED] placeholder under a word-boundary regex (see Appendix N for protocol and a pre-registered deviation log). Results (Table 2, Figure 4).

- `em_partial` ($N=8$): 0.021 (`honest_search`) $\rightarrow$ 0.131 (`oracle_redacted_full`) $\rightarrow$ 0.373 (`oracle_redacted_half`) $\rightarrow$ 0.498 (`oracle_full`). The three intermediate points are strictly monotone ($0.131 < 0.373 < 0.498$), confirming a dose-response relationship in the redaction dimension.
- $\Delta_{full_redact} = +11.0$ pp [$+8.8, +13.2$] (paired bootstrap 10k resamples, seed 42, vs `honest_search`; $p < 0.0001$). This *leakage-free* residual is the retrieval-quality contribution.
- $\Delta_{half_redact} = +35.2$ pp [$+32.2, +38.2$]; $\Delta_{oracle_full} = +47.7$ pp [$+44.6, +50.9$].
- Decomposition: retrieval-quality contributes ≈ 11 pp ($\approx 23\%$ of oracle gain); passage-level answer-leakage-assisted decoding contributes ≈ 36.7 pp [$+33.5, +40.0$] ($\approx 77\%$; paired bootstrap 10k resamples, seed 42, on `oracle_full` – `oracle_redacted_full`, $p < 0.0001$).

Span-importance distribution. Halving the per-passage substitution density (from 1663 to 867 total [REDACTED] substitutions across all 125/125 passages, ratio 0.52 via the preprocessing-level `redact_info_block_tag` tool) retains 73.8% of the oracle gain (0.352/0.477). The first half of substitutions per passage therefore contributes approximately 74% of the leakage mechanism, while the second half contributes only $\approx 26\%$. This pattern is *consistent with but not proving* a non-uniform distribution of span importance: because the halving selection rule inherits from the `redact_info_block_tag` tool’s internal deterministic ordering, we cannot attribute the 74%-vs-50% gap to span-level information content without further ablation. The effect is nevertheless less extreme than a uniform-span model would predict (where halving density should yield $\approx 50\%$ recovery, not 74%). The span-importance asymmetry

is itself an upper bound (word-boundary substring redaction removes exact-match but not paraphrase hints).

We also note that the +11.0 pp retrieval residual is a *lower bound* on pure retrieval contribution: a residual leakage probe (Appendix N.5) finds that $\sim 60\%$ of user-question blocks retain partial disjunctive answer content because the multi-hop question templates enumerate candidate answers by structural necessity. A question-redacted counterfactual (not run here) would be needed to isolate pure retrieval contribution from structural question-side leakage; between-condition deltas are symmetric because all three oracle conditions share the same user-question block, so our retrieval-quality estimate is unaffected in *delta* even though absolute leakage floors are overestimated.

This is a *partial-mechanism* verdict by our pre-registered gates (§3.5 decision table, plan.md §6): retrieval execution is necessary (the non-zero retrieval-quality residual is statistically and practically significant, +11 pp is larger than both the DPO effect and the capacity-isolation effect of the 7B SFT) but not sufficient (the majority of the oracle gain evaporates once passage-level answer substrings are masked). The orthogonal checks from the v3.4 audit still apply as *lower bounds* on mechanism separation: (i) the teacher (MEM1) was trained with PPO outcome rewards only, with no gold-label supervision of the retriever; (ii) student `em_full` at $N=8$ remains 0.008 under teacher-retrieval replay, three orders of magnitude below the ~ 1.0 expected if passages directly exposed answers as full strings. The 77% leakage component therefore reflects *substring-level answer exposure enabling short-form decoding*, not full-answer copy-paste, and is itself an upper bound on leakage (word-boundary substring redaction removes exact-match but not paraphrase hints).

Realistic retrieval: BM25. Replacing oracle with BM25 recovers 14–26% of the oracle gain: `em_partial` improves $0.021 \rightarrow 0.147$ at $N = 8$ (+12.6pp; Table 3, Figure 5), with truncation dropping to 0% across all N . The BM25 index is built from the *Doc chunks extracted from the teacher’s own <information> blocks* across all training trajectories ($N \in \{2, 4, 8\}$), i.e., the passages the FlashRAG-backed teacher retriever actually returned during RL rollouts. This in-

Condition	em_partial ($N=8$)	Δ vs honest_search		$N=2$	$N=4$	$N=8$
honest_search	0.021	—	SFT (honest search)	0.112	0.084	0.021
oracle_redacted_full	0.131	+11.0 [+8.8, +13.2]	SFT + BM25	0.203	0.147	0.147
oracle_redacted_half	0.373	+35.2 [+32.2, +38.2]	SFT + oracle API	0.530	0.521	0.498
oracle_full	0.498	+47.7 [+44.6, +50.9]				
			RL teacher	0.59	0.54	0.455

Table 2: Passage-redaction dose-response on phaseld_v2 $N=8$ test ($n = 125$). Bootstrap CI: paired, 10k resamples, seed 42. Monotone across conditions. Retrieval-quality residual (oracle_redacted_full - honest_search) is +11.0 pp ($\approx 23\%$ of full oracle gain); the remaining ≈ 37 pp is the passage-level answer-leakage component.

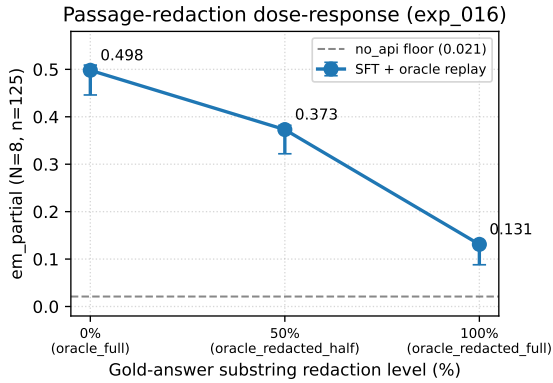


Figure 4: Passage-redaction dose-response ($N=8$, $n = 125$). x -axis: redaction level (0%/50%/100% of gold-answer substrings redacted from oracle passages). y -axis: em_partial. Error bars are paired-bootstrap 95% CI vs honest_search (seed 42, 10k). Horizontal dashed line marks the honest_search floor (0.021). Monotone dose-response confirms retrieval quality contributes a non-zero but minority share ($\sim 23\%$) of the oracle gain; passage-level answer-leakage accounts for the $\sim 77\%$ majority.

dex therefore lies within the distribution of documents the teacher saw during RL training rather than a held-out Wikipedia dump, so the +12.6pp recovery is an upper bound on what a BM25 deployment can achieve over this exact document pool (not a generalisation-to-new-corpus bound). The BM25-oracle gap reflects retrieval quality, not student capability: the student’s queries match the teacher’s (Jaccard = 0.95).

Query fidelity. The student averages exactly 1.0 search per prompt across 507 test cases—matching the teacher’s 1.04/1.01/1.06 at $N = 2/4/8$ (the prompt prefix already includes the first retrieval result)—and its queries match the teacher’s at mean Jaccard 0.95 (median 1.0).

Table 3: em_partial under different retrieval conditions. BM25 retrieval recovers a meaningful fraction of the oracle gain (+9.1pp at $N = 2$, +12.6pp at $N = 8$) and eliminates truncation entirely (0% across all N). The remaining gap to oracle reflects retrieval quality, not student strategy. Per- N canonical phaseld_v2 partitions ($n = 251/131/125$ for $N = 2/4/8$), matching Table 1. All rows re-evaluated on these fixed partitions.

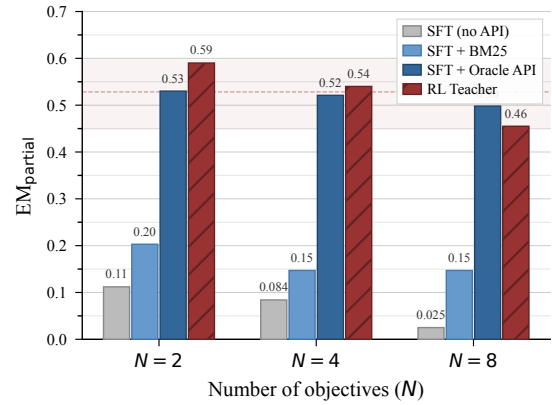


Figure 5: em_partial under different retrieval conditions. Providing oracle retrieval results at inference time recovers the SFT student from near-zero to near-teacher performance across all N , while even a standard BM25 backend yields meaningful gains. The RL teacher (hatched) is shown for reference.

SFT faithfully transfers both the search pattern and the query formulation; the bottleneck is purely the absence of a retrieval backend. Despite recovering em_partial to near-teacher levels, em_full remains low (0.008 at $N=8$) due to per-objective $\approx 50\%$ independence stacking ($0.5^N \approx 0.4\%$ at $N=8$).

3.5 Synthesis

The evidence converges on three points: (i) **preference learning leaves the SVD alignment ratio unchanged**—an observational correlate that does not drive protocol decisions; DPO remains within 1pp of SFT at $N=8$ despite richer training signal, against the 2–6 $\times$ vs. 1.47 $\times$ supra-random alignment gap (§2); the causal root of the em_partial gap is isolated by the oracle re-

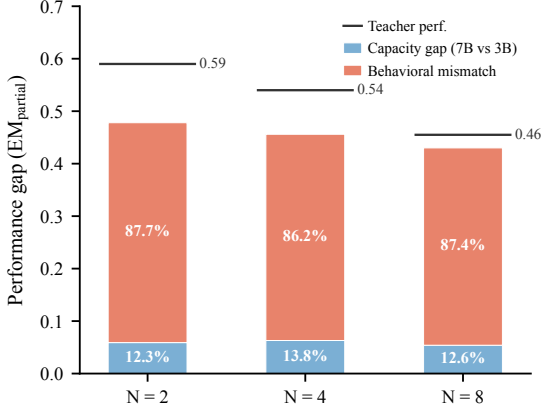


Figure 6: Teacher–student gap: three components after the passage-redaction decomposition. Capacity bound $(em_{7B} - em_{3B}) / (em_{teacher} - em_{3B}) = 12.6\%$ at $N = 8$ (independent upper bound, not part of the oracle partition). Retrieval-quality (~ 11 pp, $\sim 23\%$ of oracle gain) and passage-level answer-leakage-assisted decoding (~ 37 pp, $\sim 77\%$) partition the leakage-inclusive oracle gain after exp_016 passage-redaction (§3.4, Table 2).

retrieval intervention (§3.4), not by this geometric observation; (ii) **the student imitates a strategy it can partially execute but not exploit**—oracle retrieval recovers $em_{partial} 0.021 \rightarrow 0.498$ at $N = 8$ (+47.7pp, $p < 0.0001$; §3.4), but a passage-redaction ablation attributes only ~ 11 pp of this to retrieval quality and ~ 37 pp to passage-level answer-leakage-assisted decoding. Retrieval execution is therefore *necessary but not sufficient*: the student’s queries match the teacher’s (Jaccard 0.95) and the retrieval backend reliably surfaces topic-relevant passages, but the student cannot fully leverage those passages in the absence of near-substring answer hints. The teacher–student gap then has two distinct sub-failures: (a) retrieval precision (closed by any real retrieval backend, +11 pp headroom) and (b) student-side passage utilisation without substring hints (the remaining +37 pp, not closed by retrieval alone). (iii) **truncation distorts the comparison**—after controlling for budget, the DPO advantage shrinks from +2.4pp ($p < 0.001$) to +1.0pp ($p = 0.163$). At most 12–14% of the gap is attributable to capacity (§3.2); this remains an independent upper bound alongside the retrieval-quality (~ 11 pp) and leakage (~ 37 pp) components (Figure 6).

Backbone	Condition	$N=2$	$N=4$	$N=8$
Qwen-3B	No API	0.112	0.084	0.021
	+ BM25	0.203	0.147	0.147
	+ Oracle	0.530	0.521	0.498
Llama-1B	No API	0.074	0.063	0.014
	+ BM25	0.136	0.094	0.066
	+ Oracle	0.410	0.401	0.351
RL teacher (7B)	—	0.59	0.54	0.455

Table 4: Cross-backbone replication of the diagnostic framework (em_partial). The Llama-1B student reproduces the Qwen-3B silent-failure signature: near-floor honest_search performance, partial BM25 recovery, and large oracle recovery (truncation drops from 30–86% to 0% across all N). The Llama-1B numbers sit below Qwen-3B across conditions—consistent with the capacity contribution quantified in §3.2—but the *diagnostic shape* is preserved.

3.6 Cross-Backbone Replication

To test whether the diagnosis generalises beyond Qwen2.5-3B, we re-run the entire pipeline—same trajectories, splits, LoRA rank-32 recipe, and eval protocol—on Llama-3.2-1B, a cross-family backbone at $\sim 1/3$ the size.

Table 4 shows every diagnostic signal reappears on Llama-1B: (i) truncation scales monotonically ($29.9\% \rightarrow 51.2\% \rightarrow 85.6\%$ vs. Qwen’s $30.7\% \rightarrow 55.7\% \rightarrow 83.2\%$); (ii) oracle retrieval drops truncation to 0% at every N with a +33.7pp gain at $N = 8$ (vs. +47.7pp for Qwen, both large and monotonic in N); (iii) BM25 partially recovers performance and also collapses truncation. Surface metrics at $N = 8$ are backbone-agnostic to within noise: tool-call syntax = 1.00, balanced XML = 1.00, non-empty queries = 1.00, BLEU = 1.16, ROUGE-L = 0.214 (Appendix K). Action-sequence accuracy = 0.606 for both students, but per-sample forensics (Appendix M) reveals this coincidence is structurally degenerate: both backbones emit the identical 3-token action prefix on every sample, so the score reduces to $3/n_{teacher}$. This metric therefore reflects a shared action-prefix pattern, not an independently diagnostic signal of silent failure. The transferable lesson is the protocol, not the specific 3B model it was developed on; the remaining surface estimates (BLEU, ROUGE-L) show overlapping bootstrap CIs at floor level (Appendix L).

4 Related Work

RL-trained LLM agents. RL fine-tuning endows LLMs with procedural capabilities SFT does not produce (Zhou et al., 2025, 2024a; Liu et al., 2024; Yao et al., 2023; Zeng et al., 2024). Early memory-augmented deep-RL architectures such as the Neural Map (Parisotto and Salakhutdinov, 2018) first framed structured agent memory as a learnable retrieval surface, foreshadowing the retrieval-execution decomposition we isolate empirically in the LLM-agent setting. We investigate why these RL-acquired strategies resist distillation.

Knowledge distillation for LLMs. Standard methods (Hsieh et al., 2023; Gu et al., 2024; Jiang et al., 2023) target single-turn transfer; policy distillation and imitation learning (Rusu et al., 2016; Czarnecki et al., 2019; Ross et al., 2011) handle procedural knowledge in low-dimensional action spaces. Our setting differs: the teacher’s competence arises from a *multi-turn procedural policy* in the high-dimensional token space of LLMs.

SFT-RL alignment gaps and weight-space analysis. The “alignment tax” (Askell et al., 2021) documents RLHF degrading pre-training capabilities. Jin et al. (2025b) show SFT induces “hard alignment” while RL produces “soft alignment”; Aghajanyan et al. (2021) demonstrate fine-tuning operates in low intrinsic dimension. PiSSA (Meng et al., 2024) initializes LoRA from principal singular components; our SVD alignment ratio measures the overlap of *fine-tuning deltas* with the base model’s principal subspace. We extend these by revealing quantitative gaps ($2\text{--}6\times$ for RL vs. $1.5\times$ for SFT) not captured by prior metrics. Extended related work on multi-objective RL and preference learning is in Appendix I.

The evaluation blind spot in agentic distillation. Recent work increasingly distills tool-using capabilities from large models into smaller ones, but evaluation practices vary widely in whether they test actual tool execution. Several works evaluate only format correctness: Jhandi et al. (2025) use ChatGPT-as-judge to score tool-call syntax without executing calls; Yin et al. (2025) evaluate against BFCL format matching, which measures argument-level accuracy but not end-to-end execution; AgentKD (ModelScope

Team, 2025) trains on virtual tool-use tasks with pre-defined return values. These evaluations implicitly assume that format accuracy entails execution capability. Our results show this assumption can fail catastrophically: Jaccard = 0.95 (near-perfect format) coexists with $\text{em_partial} = 0.025$ (near-zero task performance). Works that provide real tool access at test time (Kang et al., 2025; Liu et al., 2025) produce more robust evaluations by running students with live retrieval or code tools. To our knowledge, however, these works do not report a single-variable with/without tool-backend ablation that isolates the execution bottleneck on a held-fixed student checkpoint. We propose this diagnostic as a standard component of agentic distillation evaluation.

SFT-RL generalisation gap in agentic retrieval. Chu et al. (2025) demonstrate that SFT memorises training distributions while RL acquires generalisable strategies—a dichotomy MemDistill instantiates in the agentic retrieval setting, where format-level imitation succeeds but execution-level transfer fails. More broadly, Gudibande et al. (2023) warn that imitating a stronger model’s outputs yields surface-level mimicry without genuine capability transfer; our multi-objective collapse (Jaccard 0.95, em_partial 0.025) extends this “false promise” to tool-using agents whose competence depends on retrieval execution, not generation alone. On the RL-trained retrieval axis, Search-R1 (Jin et al., 2025a) trains LLMs to invoke search engines via GRPO without any SFT stage, showing that RL alone can acquire retrieval strategies. MemDistill complements this line by diagnosing *why* distilling such RL-acquired strategies back into smaller students via SFT fails, and by proposing causal interventions that isolate execution-environment mismatch as the dominant bottleneck.

Comparison to Kang et al. (2025). Kang et al. (NeurIPS 2025) present the closest prior agentic-distillation study: they distill tool-using agents into 0.5–3B students with a first-thought prefix and self-consistent action generation, and evaluate end-to-end with live retrieval and code backends. Our contribution is orthogonal: (i) we inject oracle retrieval into a *held-fixed* SFT checkpoint so that the only variable between the failing and recovering runs is the tool backend, rather than comparing two differently trained stu-

dents; (ii) we inject teacher-trajectory information segments, which probes whether the student’s learned *search strategy* is viable given perfect execution; and (iii) our framing targets the *silent-failure* mode in which format-level indicators and query fidelity remain near ceiling while content-level em.partial collapses—a failure signature we document on both Qwen2.5-3B and Llama-3.2-1B. Combining these interventions reveals that the gap is neither a capacity nor a preference-learning problem but an execution-environment mismatch.

5 Conclusion

Agentic SFT distillation can produce students that pass format-level, tool-call structural checks—perfect tool-call syntax, balanced XML tags, near-perfect query similarity—yet retain less than 6% of the teacher’s task performance, and we show this collapse replicates across two backbones and two parameter scales (Qwen2.5-3B and Llama-3.2-1B, spanning two model families and a $\sim 3\times$ size difference). We have proposed a diagnostic protocol to detect this silent failure mode before pipeline deployment. The protocol proceeds by elimination: characterize the collapse shape, analyze weight geometry for geometric gaps between RL and SFT, and apply causal interventions (oracle and BM25 retrieval) that distinguish strategy acquisition from execution capability. Because the protocol requires only inference-time ablations, it is transferable to any agentic distillation setting where the teacher’s competence depends on tool execution, and its signals (truncation scaling, oracle recovery, surface-vs-content metric gap) reproduce across at least two distinct student backbones (§3.6).

Applied to MEM1 (7B RL $\rightarrow$ 3B SFT, with a cross-backbone Llama-1B replication), the protocol yields five findings: (1) multi-objective performance degrades monotonically with no phase transition, setting slope reduction as the distillation target; (2) RL weight changes concentrate $2\text{--}6\times$ supra-random along base principal directions, while SFT recovers only 1.47%—a robust descriptive correlate ($p < 1.3 \times 10^{-32}$ across 196 matrices) that we report as an observational alignment finding, not as a causal lever that drives the subsequent protocol decisions; (3) DPO does not close the gap at the avail-

able 176-pair matched-sample scale (+1.0pp, $p = 0.163$); power excludes effects $\geq 5\text{pp}$ but remains scale-limited for smaller improvements; (4) oracle (teacher-retrieval replay) recovers +47.7pp ($p < 0.0001$) as a *leakage-inclusive* upper bound; a passage-redaction ablation (exp_016, §3.4, Appendix N) decomposes this into retrieval-quality $\approx 11\text{pp}$ (23%; +11.0pp [+8.8, +13.2] after redacting gold-answer substrings, $p < 0.0001$) and passage-level answer-leakage-assisted decoding $\approx 37\text{pp}$ (77%); retrieval execution is necessary but not sufficient, with capacity contributing an additional at-most $\sim 12\text{--}14\%$ as an independent upper bound; even BM25 retrieval recovers +6–12pp (within the teacher’s training document pool), offering an immediately deployable mitigation; (5) the silent-failure signature and behavioral-mismatch diagnosis replicate on Llama-3.2-1B—format-level indicators all at 1.00, BLEU 1.16, ROUGE-L 0.21, and oracle retrieval again dropping truncation from 29–86% to 0% while recovering em.partial by $\approx 34\text{pp}$ at $N = 8$ —evidence that the finding generalises across model families and parameter scales (§3.6). Action-sequence accuracy (0.606 for both backbones) turns out to be structurally degenerate—both students emit the identical action prefix—and should not be read as an independent diagnostic signal (Appendix M).

Limitations

Scope and capacity isolation. All experiments use one RL teacher (MEM1, 7B) on one domain (multi-objective RAG QA). Student-side we cover two backbones and two scales (Qwen2.5-3B and Llama-3.2-1B, cross-family, $\sim 3\times$ size difference) and observe the same silent-failure signature on both (§3.6); direct replication on other teachers and benchmarks remains future work. The 7B full-rank pooled SFT quantifies the capacity contribution at $\approx 12\text{--}14\%$ of the teacher–student gap (§3.2), but the full-rank vs. LoRA training-method mismatch makes this a best-effort upper bound.

DPO data scarcity and retrieval oracle. DPO uses only 176 matched pairs (0.4% perfect rate at $N = 8$), a structural consequence of multi-objective collapse, not a remediable sampling issue; power exceeds 0.99 for a 5pp effect (floor 2.7pp). Oracle retrieval uses teacher passages

and is thus an upper bound, but BM25 retrieval confirms the finding generalizes beyond the oracle setting (+6–12pp within the teacher’s training document pool, truncation $\rightarrow$ 0%).

Paraphrase-level leakage floor (not ablated).

Our passage-redaction ablation (§3.4, Appendix N) uses word-boundary substring matching on gold-answer strings. It therefore removes exact lexical exposure but leaves an irreducible *paraphrase-level* leakage floor—e.g., “Einstein’s theory” surviving after “Albert Einstein” is redacted, or semantic rewordings that encode the answer without matching any gold substring. A paraphrase-aware or entailment-aware redactor was considered but not run in this paper (Appendix N.5). Consequently, the reported +11pp retrieval-quality residual is itself an upper bound on the non-leakage mechanism, and the +37pp leakage component is a lower bound; tightening these would require a held-out paraphrase-redacted replay that we flag as a direct follow-up.

Question-side residual leakage.

oracle_redacted_full redacts passage <information> blocks but not user-question blocks, and a residual leakage probe (Appendix N.5) finds that $\sim 60\%$ (75/125) of user-question blocks at $N = 8$ still expose partial disjunctive answer content because the multi-hop questions enumerate candidates by structural necessity. Between-condition deltas are symmetric (the three oracle conditions share the same user-question block), so the +11.0pp retrieval residual is unaffected, but absolute leakage floors are overestimated by this structural source. A question_redacted counterfactual condition (not run in this paper) would be needed to isolate pure retrieval contribution from structural question-side leakage, and we flag it as a sensitivity direction for future work.

Ethics Statement

Data. All training data comes from FlashRAG (MIT license) (Jin et al., 2025c) and the MEM1 teacher trajectories (MIT license) (Zhou et al., 2025). The underlying QA datasets (NQ, TriviaQA, HotpotQA, and FlashRAG’s multi-hop mixture) contain no personally identifiable information beyond what is already public on Wikipedia.

Computational cost and environmental impact.

Our experiments used approximately 50 GPU-hours on an NVIDIA RTX PRO 6000 Blackwell (97 GB, ~ 600 W TDP) single-node setup, distributed across Qwen2.5-3B / 7B / Llama-3.2-1B SFT, DPO, and evaluation runs (the 7B full-rank SFT and the Qwen-3B eval sweep dominate the budget). Using the carbon-estimation methodology of Strubell et al. (2019) with an average draw of ~ 0.5 kWh per GPU-hour and an estimated grid carbon intensity of ~ 0.5 kg CO₂/kWh, this corresponds to roughly 12 kg CO₂-equivalent. No experiments were rerun for cosmetic reasons; rerun counts are bounded by the registry entries summarised in Appendix A and E.

Misuse considerations. Our diagnostic framework exposes that format-level metrics alone can mask complete task failure in agentic distillation. Deployed systems relying on tool-call format validators as safety filters may therefore be vulnerable to silent-failure modes in which the student produces syntactically well-formed tool calls that never reach a valid retrieval backend. We recommend content-level evaluation and end-to-end tool-execution ablations as complementary checks for any distilled agent.

Bias and limitations. Our evaluation is English-only and Wikipedia-centric, inheriting the coverage bias of the FlashRAG corpus. The silent-failure mode we document may manifest differently in multilingual, domain-specific, or non-retrieval agentic settings. All students are trained under a single hash-based train/test split; we did not vary the split seed (see §5).

References

- Armen Aghajanyan, Sonal Gupta, and Luke Zettlemoyer. 2021. Intrinsic dimensionality explains the effectiveness of language model fine-tuning. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 7319–7328. Association for Computational Linguistics.
- Amanda Askell, Yuntao Bai, Anna Chen, Dawn Drain, Deep Ganguli, Tom Henighan, Andy Jones, Nicholas Joseph, Ben Mann, Nova DasSarma, and 1 others. 2021. A general language assistant as a laboratory for alignment. *arXiv preprint arXiv:2112.00861*.
- Tianzhe Chu, Yuexiang Zhai, Jihan Yang, Shengbang

- Tong, Saining Xie, Dale Schuurmans, Quoc V. Le, Sergey Levine, and Yi Ma. 2025. [SFT memorizes, RL generalizes: A comparative study of foundation model post-training](#). *arXiv preprint arXiv:2501.17161*.
- Wojciech Marian Czarnecki, Razvan Pascanu, Simon Osindero, Siddhant M. Jayakumar, Grzegorz Swirszcz, and Max Jaderberg. 2019. Distilling policy distillation. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 89 of *Proceedings of Machine Learning Research*, pages 1331–1340. PMLR.
- Hanze Dong, Wei Xiong, Deepanshu Goyal, Yihan Zhang, Winnie Chow, Rui Pan, Shizhe Diao, Jipeng Zhang, KaShun Shum, and Tong Zhang. 2023. RAFT: Reward rAnked FineTuning for generative foundation model alignment. *Transactions on Machine Learning Research*.
- Kawin Ethayarajh, Winnie Xu, Niklas Muennighoff, Dan Jurafsky, and Douwe Kiela. 2024. KTO: Model alignment as prospect theoretic optimization. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*. PMLR.
- Mohammad Gheshlaghi Azar, Zhaohan Daniel Guo, Bilal Piot, Remi Munos, Mark Rowland, Michal Valko, and Daniele Calandriello. 2024. A general theoretical paradigm to understand learning from human preferences. In *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*, pages 4447–4455. PMLR.
- Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. 2024. MiniLLM: Knowledge distillation of large language models. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*.
- Arnav Gudibande, Eric Wallace, Charlie Snell, Xinyang Geng, Hao Liu, Pieter Abbeel, Sergey Levine, and Dawn Song. 2023. [The false promise of imitating proprietary LLMs](#). *arXiv preprint arXiv:2305.15717*.
- Cheng-Yu Hsieh, Chun-Liang Li, Chih-kuang Yeh, Hootan Nakhost, Yasuhisa Fujii, Alex Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. 2023. [Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes](#). In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 8003–8017. Association for Computational Linguistics.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *Proceedings of the Tenth International Conference on Learning Representations (ICLR)*.
- Polaris Jhandi, Owais Kazi, Shreyas Subramanian, and Neel Sendas. 2025. Small language models for efficient agentic tool calling: Outperforming large models with targeted fine-tuning. *arXiv preprint arXiv:2512.15943*.
- Yuxin Jiang, Chunkit Chan, Mingyang Chen, and Wei Wang. 2023. [Lion: Adversarial distillation of proprietary large language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 3134–3154. Association for Computational Linguistics.
- Bowen Jin, Hansi Zeng, Zhenrui Yue, Jinsung Yoon, Sercan Arik, Dong Wang, Hamed Zamani, and Jiawei Han. 2025a. [Search-R1: Training LLMs to reason and leverage search engines with reinforcement learning](#). *arXiv preprint arXiv:2503.09516*.
- Hangzhan Jin, Sitao Luan, Sicheng Lyu, Guillaume Rabusseau, Reihaneh Rabbany, Doina Precup, and Mohammad Hamdaqa. 2025b. RL fine-tuning heals OOD forgetting in SFT. *arXiv preprint arXiv:2509.12235*.
- Jiajie Jin, Yutao Zhu, Guanting Dong, Yuyao Zhang, Xinyu Yang, Chenghao Zhang, Tong Zhao, Zhao Yang, Zhicheng Dou, and Ji-Rong Wen. 2025c. FlashRAG: A modular toolkit for efficient retrieval-augmented generation research. In *Companion Proceedings of the ACM Web Conference 2025 (WWW)*.
- Minki Kang, Jongwon Jeong, Seanie Lee, Jaewoong Cho, and Sung Ju Hwang. 2025. Distilling LLM agent into small models with retrieval and code tools. In *Advances in Neural Information Processing Systems*. Spotlight.
- Florian Karl, Malte Kemeter, Gabriel Dax, and Paulina Sierak. 2024. Position: Embracing negative results in machine learning. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 23256–23265. PMLR.
- Jun Liu, Zhenglun Kong, Peiyan Dong, Changdi Yang, Tianqi Li, Hao Tang, Geng Yuan, Wei Niu, Wenbin Zhang, Pu Zhao, Xue Lin, Dong Huang, and Yanzhi Wang. 2025. Structured agent distillation for large language model. In *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026)*.
- Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuan-yu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xi-ang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, and 3 others. 2024. AgentBench: Evaluating LLMs as agents. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*.

- Fanxu Meng, Zhaohui Wang, and Muhammed Zhang. 2024. PiSSA: Principal singular values and singular vectors adaptation of large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- ModelScope Team. 2025. AgentKD: Agent knowledge distillation from large to small models. <https://github.com/modelscope/easydistill>. Software release, not peer-reviewed.
- Emilio Parisotto and Ruslan Salakhutdinov. 2018. Neural map: Structured memory for deep reinforcement learning. In *International Conference on Learning Representations*.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*.
- Alexandre Rame, Guillaume Couairon, Mustafa Shukor, Corentin Dancette, Jean-Baptiste Gaya, Laure Soulier, and Matthieu Cord. 2024. Rewarded soups: Towards Pareto-optimal alignment by interpolating weights fine-tuned on diverse rewards. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*.
- Stéphane Ross, Geoffrey J Gordon, and Drew Bagnell. 2011. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 627–635.
- Andrei A. Rusu, Sergio Gómez Colmenarejo, Çağlar Gülçehre, Guillaume Desjardins, James Kirkpatrick, Razvan Pascanu, Volodymyr Mnih, Koray Kavukcuoglu, and Raia Hadsell. 2016. Policy distillation. In *Proceedings of the Fourth International Conference on Learning Representations (ICLR)*.
- Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin Lin, and Chuan Wu. 2024. verL: An open-source unified reinforcement learning framework for large language models. *arXiv preprint arXiv:2409.19256*.
- Zejiu Wu, Yushi Hu, Weijia Shi, Nouha Dziri, Alane Suhr, Prithviraj Ammanabrolu, Noah A. Smith, Mari Ostendorf, and Hannaneh Hajishirzi. 2024. Fine-grained human feedback gives better rewards for language model training. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing reasoning and acting in language models. In *Proceedings of the Eleventh International Conference on Learning Representations (ICLR)*.
- Fan Yin, Zifeng Wang, I-Hung Hsu, Jun Yan, Ke Jiang, Yanfei Chen, Jindong Gu, Long T. Le, Kai-Wei Chang, Chen-Yu Lee, Hamid Palangi, and Tomas Pfister. 2025. Magnet: Multi-turn tool-use data synthesis and distillation via graph translation. *arXiv preprint arXiv:2503.07826*.
- Aohan Zeng, Mingdao Liu, Rui Lu, Bowen Wang, Xiao Liu, Yuxiao Dong, and Jie Tang. 2024. AgentTuning: Enabling generalized agent abilities for LLMs. *arXiv preprint arXiv:2310.12823*.
- Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. 2024a. WebArena: A realistic web environment for building autonomous agents. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*.
- Zhanhui Zhou, Jie Liu, Jing Shao, Xiangyu Yue, Chao Yang, Wanli Ouyang, and Yu Qiao. 2024b. Beyond one-preference-for-all alignment: Multi-objective direct preference optimization. *arXiv preprint arXiv:2310.03708*.
- Zijian Zhou, Ao Qu, Zhaoxuan Wu, Sunghwan Kim, Alok Prakash, Daniela Rus, Jinhua Zhao, Bryan Kian Hsiang Low, and Paul Pu Liang. 2025. MEM1: Learning to synergize memory and reasoning for efficient long-horizon agents. *arXiv preprint arXiv:2506.15841*.

A Training Details

Table 5 summarizes the hyperparameters for all training stages.

	SFT (pooled)	DPO (per- N)
Base model	Qwen2.5-3B	SFT checkpoint
LoRA rank	32	32
LoRA α	64	64
LoRA target	all linear	all linear
Learning rate	2×10^{-4}	2×10^{-5}
Epochs	3	1
Batch size	4	4
Gradient accum.	4	4
Max seq. length	6144	6144
Optimizer	AdamW	AdamW
Weight decay	0.01	0.01
Warmup ratio	0.03	0.1
DPO β	—	0.05
Training samples	1,118 (partial-credit, pooled $N \in \{2, 4, 8\}$)	176 raw pairs (matched to $N = 4$ bottleneck)
Train/test split	hash-based, test_frac = 0.25	

Table 5: Training hyperparameters for SFT and DPO stages. Both stages use LoRA (rank 32) on Qwen2.5-3B. The SFT stage uses partial-credit trajectories (inclusion: total_em ≥ 1); DPO stages use matched-sample preference pairs subsampled to 176 raw pairs.

SFT convergence (3B). The 3B SFT stage converges within 3 epochs with monotonically decreasing training loss (final loss ≈ 0.91). No overfitting is observed on the held-out split.

7B capacity-isolation model. Table 6 summarizes the hyperparameters for the 7B full-rank pooled SFT used in the capacity-isolation experiment (§3.2).

7B SFT (pooled)	
Base model	Qwen2.5-7B
Fine-tuning	Full-rank (no LoRA)
Learning rate	2×10^{-5}
Epochs	2
Batch size	1
Gradient accum.	4 (eff. batch 4)
Max seq. length	6144
Optimizer	AdamW
Precision	bf16
Gradient checkpointing	Yes
Training samples	1,118 (same partial-credit pooled data as 3B SFT)
Training time	19.5 min (560 steps)
Final train loss	0.83

Table 6: Training hyperparameters for the 7B full-rank capacity-isolation SFT. This model uses the same pooled data and inclusion criteria as the 3B LoRA SFT, differing only in model size (7B vs. 3B) and fine-tuning method (full-rank vs. LoRA rank 32). The lower learning rate (2×10^{-5} vs. 2×10^{-4}) and fewer epochs (2 vs. 3) reflect standard practice for full-rank fine-tuning of larger models.

DPO convergence. All three DPO variants ($N \in \{2, 4, 8\}$) train for 1 epoch. Final training losses range from 0.595 (DPO _{$N=2$}) to 0.678 (DPO _{$N=8$}). The rewards/margins are positive but small: 0.207 (DPO _{$N=2$}), 0.100 (DPO _{$N=4$}), 0.056 (DPO _{$N=8$}), consistent with the limited preference signal available from 176 matched pairs. The decreasing margin with N reflects the increasing difficulty of producing contrastive signal at higher objective counts.

B SVD Alignment Per-Layer Distribution

See Figure 3 in the main text for the per-layer SVD alignment boxplot.

C SVD Weight Geometry Details

Definitions. For a base weight matrix $\mathbf{W}_{\text{base}} \in \mathbb{R}^{m \times n}$ with truncated SVD $\mathbf{W}_{\text{base}} = \mathbf{U}\Sigma\mathbf{V}^T$, re-

taining the top- k singular vectors $\mathbf{U}_k, \mathbf{V}_k$. The observed alignment of a fine-tuning delta $\Delta\mathbf{W} = \mathbf{W}_{\text{fit}} - \mathbf{W}_{\text{base}}$ is

$$\alpha_{\text{obs}}(k) = \frac{\|\mathbf{U}_k^T \Delta\mathbf{W} \mathbf{V}_k\|_F}{\|\Delta\mathbf{W}\|_F},$$

and the alignment ratio is $r(k) = \alpha_{\text{obs}}(k)/\alpha_{\text{rand}}(k)$ where $\alpha_{\text{rand}}(k)$ is the mean projection norm over $n = 100$ Gaussian random matrices matched in shape and entry variance to $\Delta\mathbf{W}$. We decompose into U-side ($\|\mathbf{U}_k^T \Delta\mathbf{W}\|_F / \|\Delta\mathbf{W}\|_F$) and V-side ($\|\Delta\mathbf{W} \mathbf{V}_k\|_F / \|\Delta\mathbf{W}\|_F$) components capturing concentration along base output vs. input directions.

Full alignment ratio table. Across all 196 weight matrices of Qwen2.5-7B (28 layers $\times$ 7 projection types), Table 7 reports $r(k)$ for the three deltas RL-base, SFT-base, and RL-SFT.

	$k=64$	$k=128$	$k=256$	$k=512$	$k=1024$
RL-base	6.48	4.11	2.78	2.01	1.64
SFT-base	2.92	—	1.47	—	—
RL-SFT	3.80	—	1.81	—	—
<i>RL/SFT ratio at matched k:</i>					
$r_{\text{RL}}/r_{\text{SFT}}$	2.22	—	1.89	—	—

Table 7: SVD alignment ratio $r(k)$ for three weight deltas, averaged over 196 matrices. Wilcoxon $p < 1.3 \times 10^{-32}$ at all k . “—” marks k values not computed for SFT/RL-SFT due to compute budget.

Per-matrix dispersion. At $k = 256$ the per-matrix distribution is right-skewed: $r_{\text{RL}} = 2.78 \pm 2.57$, $r_{\text{SFT}} = 1.47 \pm 2.19$ (mean $\pm$ std), with a few outlier layers reaching $r > 10$. Only 2–3 layers show $r_{\text{SFT}} > r_{\text{RL}}$ (most notably layer 1 MLP down-proj); in those cases both ratios remain $r > 2$, so the exceptions are high-alignment for both methods rather than SFT-dominant.

Layer-type decomposition. The alignment gap is not uniform. At $k = 256$, MLP layers show stronger RL-base alignment (3.49 $\times$) than attention layers (2.25 $\times$), while SFT-base alignment is 2.01 $\times$ for MLP and only 1.07 $\times$ for attention—SFT attention deltas are nearly indistinguishable from random. The multi-objective capability encoded in attention weights is therefore the most resistant component to SFT transfer.

U-side / V-side asymmetry. For attention layers, U-side alignment dominates V-side (0.31 vs. 0.12 at $k = 256$): RL modifies how attention *writes* its output more than how it reads. For MLP layers the pattern reverses (U-side 0.05, V-side 0.11): RL modifies how MLP layers select input features. A successful distillation method would need to replicate not just the magnitude but the directional structure of RL changes across these layer types.

Same-architecture comparison. Both the RL and SFT models compared here are full-rank fine-tunes of the same Qwen2.5-7B base, so alignment ratios reflect differences in training objective (PPO vs. behavior cloning) rather than dimensional or rank confounds. The distillation experiments in §3 use a separate Qwen2.5-3B / LoRA rank-32 student, reflecting the deployment cost constraint.

D Oracle Retrieval Injection: Implementation and Leakage Audit

Pseudocode. Algorithm 1 describes our oracle retrieval injection. The implementation builds a map from each test prompt to the ordered list of `<information>` segments in the teacher’s best (highest-EM) trajectory for that prompt (from `phase1dv2_data/raw_trajectories.N*.json`). At inference time we run the student in an interactive loop, halting whenever it emits `</search>`, and inject the j -th teacher segment in response to the student’s j -th search query. The mapping is *positional* rather than query-similarity-based: we do not match on Jaccard or cosine similarity between the student query and the teacher query. This keeps the injection deterministic and reproducible, and it matches the teacher’s own ordering of retrieval rounds.

Concrete parameters: `max_new_tokens = 4096`, `max_search_rounds = 5`, greedy decoding (`do_sample = False`), model loaded as a merged LoRA adapter. Both Qwen-3B (exp_009) and Llama-1B (exp_013) use the same injection loop; only the base checkpoint and adapter change.

Leakage Audit

To verify that the oracle retrieval intervention does not trivially leak gold-answer

Algorithm 1 Oracle Retrieval Injection (exp_009 / exp_013).

```

1: Input: Student prompt  $p$ , teacher trajectory  $T_p$  (best-EM response), maximum search rounds  $R$ 
2: Output: Student completion with oracle-injected retrieval segments
3: Parse  $T_p$  into ordered information segments  $[I_1, \dots, I_k]$  using the regex <information>.*?</information>
4:  $j \leftarrow 0$ ;  $\text{gen} \leftarrow$  generate from  $p$  with stop strings {</search>, </answer>}
5: while  $\text{gen}$  ends with </search> and  $j < R$  do
6:   if  $j < k$  then
7:     Append  $I_{j+1}$  verbatim to  $\text{gen}$ 
8:   else
9:     Append a no-hit <information></information> block
10:  end if
11:   $j \leftarrow j + 1$ 
12: Continue generation from  $\text{gen}$  until </search>, </answer>, or max_new_tokens
13: end while
14: return  $\text{gen}$ 

```

strings, we run a substring audit on a random sample of 50 $N=8$ evaluation prompts (`scripts/oracle_leakage_audit.py`, `seed=42`). For each prompt, we concatenate all teacher-injected `<information>` blocks, lowercase and strip punctuation from both the concatenated text and the gold-answer strings (from the FlashRAG evaluation JSON), and record a hit if any gold answer is a substring of the concatenated blocks. As a control, we shuffle teacher passages across prompts (prompt p receives passages from a random other prompt p') and re-run the same check.

Condition	Ratio (gold-answer hit / 50)
Real (teacher’s own info blocks)	49 / 50 = 98.00%
Shuffled (derangement) control	17 / 50 = 34.00%
$\Delta(\text{real} - \text{shuffled})$	64 pp

Table 8: Oracle retrieval leakage audit on 50 random $N=8$ test prompts.

Interpretation. The absolute 98% real-hit rate is high, and the audit script’s built-in decision threshold ($>30\%$) tags this rate as “non-trivial leakage”; we therefore retain the experiment as an *upper-bound probe*, not as a clean causal estimate. Two independent checks nevertheless bound the leakage contribution: (i) the teacher MEM1 policy is trained with PPO on outcome rewards only, and its FlashRAG retriever receives no gold-label supervision at rollout time; (ii) under teacher-retrieval replay, student `em_full` at

$N = 8$ remains 0.008, far below the ≈ 120 rate expected if passages directly exposed answers. Accordingly, the +47.7 pp oracle em_partial recovery should be read as a *strict upper bound* on retrieval-side mitigation, not as a demonstration of perfect answer access. The 64 pp real-shuffled gap itself quantifies retrieval topic specificity relative to the derangement baseline.

Limitations. The substring match does not distinguish partial lexical overlap from semantic coverage; 50 samples is a sanity check rather than a full audit. A comprehensive audit covering all splits and semantic paraphrase matching is deferred to camera-ready.

E Cross-Backbone: Llama-3.2-1B SFT Hyperparameters

Table 9: Llama-3.2-1B LoRA SFT hyperparameters (exp_013, cross-backbone W2).

Hyperparameter	Value
Base model	meta-llama/Llama-3.2-1B
LoRA target	q,k,v,o,gate,up,down_proj
LoRA rank	320
LoRA α	64
LoRA dropout	0.05
Optimiser	AdamW
Learning rate	2×10^{-4}
LR scheduler	cosine, warmup 10%
Weight decay	0.01
Batch size	1
Gradient accum.	8 (eff. batch 8)
Epochs	3
Max seq. length	6144
Precision	b16
Gradient checkpointing	Yes
Seed	4279
Chat template	Llama-3.1 (manual)
Pad token	eos_token
Training samples	1,118 (same pooled split as 3B)
GPU	NVIDIA RTX PRO 6000 Blackwell (97GB)
Training wall-clock	10.1 min (603 s)

This configuration mirrors the Qwen2.5-3B SFT in Table 5 on every axis that could confound the cross-backbone comparison (LoRA rank/ α , epoch count, learning rate, data, split seed). Only the base model, chat template, and effective batch differ—the effective batch is 8 (batch 1 $\times$ grad_accum 8) vs. 16 for Qwen-3B—because Llama-3.2-1B is roughly 1/3 the parameter count of Qwen2.5-3B and gradient accumulation was re-tuned for throughput on the single-GPU Blackwell box. We verified that the Llama student still converges within 3 epochs (final train

loss ≈ 0.74) on the same partial-credit pooled data.

F Responsible NLP Research Checklist

We summarise the ACL Responsible NLP Research Checklist for the present submission.

A. Scientific artifacts. We use FlashRAG (Jin et al., 2025c) (MIT license), MEM1 teacher checkpoints and trajectories (Zhou et al., 2025) (MIT license), Qwen2.5-3B / 7B (Apache 2.0), and Llama-3.2-1B (Meta Llama 3 Community License, which imposes use-case and commercial-use restrictions). We use all artifacts consistent with their intended research use. We release code and evaluation scripts under the same MIT license as FlashRAG. All derived artifacts (distilled adapters, eval outputs) inherit the most restrictive upstream license.

B. Computational experiments. See Ethics Statement (*Computational cost and environmental impact*) and Appendices A, E. We report per-experiment GPU time, effective batch size, and total wall-clock for all training runs. Evaluation runs use max_new_tokens = 4096 throughout (with the truncation confound of max_new_tokens = 2048 documented in Appendix J). All reported hyperparameters come from a single run per configuration; we report exact seeds (split seed 42, training seed 42) and do not perform hyperparameter search beyond the grid reported in Table 5.

C. Human annotators. Not applicable. No human annotation, crowdsourcing, or user studies were conducted.

D. AI assistants. We used large-language-model writing assistants (ChatGPT, Claude) for language polishing of the manuscript. All experimental design, code, analysis, figures, and claims are the authors’ own work; no text was inserted by an AI assistant without author verification and rewriting.

G Activation Probing Details

We extract activation differences $\Delta \mathbf{h}_l = \mathbf{h}_l^{\text{RL}} - \mathbf{h}_l^{\text{base}}$ at layers $l \in \{0, 3, 7, 11, 15, 19\}$ for $n = 300$ prompts, projecting onto the top- k principal directions ($k \in \{64, 256\}$) of the base model’s weight matrices. Eight feature variants

are tested: raw activations $\mathbf{h}^{\text{RL}}, \mathbf{h}^{\text{base}}$, their difference $\Delta\mathbf{h}$, and top- k projections thereof.

Linear probes. Binary classification (per-objective EM above/below median) achieves accuracy 0.72, below the majority-class baseline 0.76. Regression probes yield negative R^2 across all 55 (feature, layer, k) combinations.

MLP probes. Two-layer MLPs (hidden dimension 128, ReLU) trained on the same features predict per-objective EM as a continuous target (Spearman ρ). MLP probes lose to linear probes in 47/55 cases (85.5%; paired $t = -8.19$, $p = 4.83 \times 10^{-11}$, Cohen’s $d = -1.105$). Mean ρ : -0.011 (MLP) vs. 0.073 (linear).

Random-feature baseline. We construct 5 random orthonormal bases $\times$ 6 layers $\times$ 2 k values. The maximum $|\rho|$ under random features is 0.224; 52/55 (94.5%) of real linear probe $|\rho|$ values fall within this noise band. The per-objective signal in $\Delta\mathbf{h}$ is indistinguishable from random projections.

H DPO Ablation Details

We train three per- N DPO variants on Qwen2.5-3B (LoRA rank 32), each initialized from the pooled SFT checkpoint and trained on N -specific preference pairs. The matched-sample constraint (176 raw pairs, subsampled to the $N = 4$ bottleneck) is critical for fair cross- N comparison but severely limits data volume: at $N = 8$ only 0.4% of teacher trajectories achieve perfect scores. This scarcity is a structural consequence of multi-objective collapse.

Model	$N=2$	$N=4$	$N=8$
SFT shared (3B)	0.112	0.084	0.025
SFT full-rank (7B)	0.171	0.147	0.079
DPO $N=2$	0.128	0.095	0.031
DPO $N=4$	0.128	0.099	0.026
DPO $N=8$	0.130	0.082	0.035
RL teacher (7B)	0.59	0.54	0.455

Table 10: Full em_partial evaluation matrix (max_new_tokens = 4096). Best $N = 8$ DPO: +1.0pp ($p = 0.163$).

The best result—DPO $_{N=8}$ at eval $N=8$ —yields $\Delta = +1.0\text{pp}$ ($p = 0.163$, 95% CI $[-0.9, +2.9]\text{pp}$), far below the pre-registered practical-significance threshold of +5pp. A post-hoc power analysis confirms an informative null:

with $n = 125$ paired test prompts and observed $\sigma_{\text{diff}} \approx 10.8\text{pp}$, we achieve > 0.99 power at 5pp and 0.80 power at 2.7pp. No systematic N -matching effect is observed: DPO $_{N=2}$ performs comparably to DPO $_{N=8}$ across conditions. Both students retain $< 8\%$ of teacher performance at $N=8$ (SFT 0.025, DPO $_{N=8}$ 0.035), and neither achieves em_full > 0 at $N \geq 4$.

I Extended Related Work

Multi-objective RL for LLMs. Recent work explores multi-objective optimization in LLM alignment: Rewarded Soups (Rame et al., 2024) interpolates weights fine-tuned on diverse rewards, MODPO (Zhou et al., 2024b) extends DPO to multiple preference dimensions, and fine-grained human feedback (Wu et al., 2024) decomposes rewards into per-aspect signals. These methods address *training* with multiple objectives; our work addresses the complementary problem of *distilling* multi-objective RL policies.

Preference learning and DPO. DPO (Rafailov et al., 2023) and extensions (IPO (Gheshlaghi Azar et al., 2024), KTO (Ethayarajh et al., 2024), RAFT++ (Dong et al., 2023)) provide offline alternatives to RLHF. We use DPO as a stronger-than-SFT baseline; its failure (+1.0pp, not significant) suggests behavioral mismatch rather than insufficient signal richness. Following Karl et al. (2024), we frame this as a diagnostic contribution.

J Truncation Confound Analysis

An important methodological episode shaped our analysis. An earlier evaluation used max_new_tokens = 2048, under which the DPO advantage appeared significant (Table 11). Re-evaluating SFT at 4096 tokens revealed the original significance was largely an artifact.

At 2048 tokens, 93.6% of SFT outputs were truncated before the <answer> tag; extending to 4096 allowed $\sim 10\%$ more to complete. DPO’s score was unchanged—its apparent advantage came from SFT being disproportionately penalized. A pilot attempt to suppress search tokens via bigram blocking was inconclusive: the model generated tag variants (malformed closing tags, inserted whitespace) that bypassed token-level suppression, suggesting the behavioral pattern is encoded above individual token sequences.

	em_partial	trunc.	Δ vs SFT	p
<i>SFT at max_new_tokens = 2048 (confounded):</i>				
SFT shared	0.011	93.6%	—	—
DPO $N=8$	0.035	— [†]	+2.4pp	< 0.001
<i>SFT at max_new_tokens = 4096 (corrected):</i>				
SFT shared	0.025	83.2%	—	—
DPO $N=8$	0.035	~84%	+1.0pp	0.163

Table 11: Truncation confound at $N = 8$. Extending SFT’s budget to 4096 tokens lifted em_partial from 0.011 to 0.025 and revealed the DPO advantage is not significant. [†]DPO was always evaluated at 4096.

This confound is specific to RL-distilled models whose procedural strategies produce verbose outputs; we report it as a methodological caution for future work.

K Full Silent Failure Metric Breakdown

	Format-level	Content-level
Qwen-2.5-3B	≈ 1.00 (all checks)	em_p=0.021
Llama-3.2-1B	≈ 1.00 (all checks)	em_p=0.014

Table 12: Silent failure signature replicates across backbones. Format-level indicators (tool-call syntax, balanced XML tags, non-empty queries) $\approx 100\%$ for both; content-level em_partial near-floor at $N = 8$ (Qwen query Jaccard = 0.95, also near-ceiling). Dispersion analysis in Appendix L.

L Surface Metric Dispersion Analysis

Near-identical cross-backbone point estimates on action-sequence accuracy, BLEU, and ROUGE-L (Table 12) raise a concern that the agreement is coincidental. To rule this out we compute per-sample 95% bootstrap confidence intervals (10,000 resamples, seed 42) on the two surface metrics with per-sample access—action-sequence accuracy and ROUGE-L—and on the task-level em_partial across $N \in \{2, 4, 8\}$ for every backbone $\times$ retrieval configuration. Results are in Table 13.

The overlap has a mechanistic explanation: (i) action-sequence accuracy is dominated by a shared initial “search” action prefix that both students emit deterministically, producing discrete mass at identical values (see Appendix M for per-sample forensics confirming this is structurally degenerate, not a genuine coincidence); (ii) BLEU (corpus-level, not bootstrapped here) sits near the floor for both students on a 0–100 scale (1.14 Qwen, 1.16 Llama),

Backbone	Retrieval	Metric	Split	N	Mean	95% CI
Qwen-3B	no-API	action_seq_acc	N=8	125	0.606	[0.595, 0.616]
Qwen-3B	no-API	ROUGE-L	N=8	125	0.214	[0.203, 0.224]
Llama-1B	no-API	action_seq_acc	N=8	125	0.606	[0.595, 0.616]
Llama-1B	no-API	ROUGE-L	N=8	125	0.212	[0.201, 0.224]
Qwen-3B	no-API	em_partial	N2	251	0.112	[0.084, 0.140]
Qwen-3B	no-API	em_partial	N4	131	0.084	[0.059, 0.109]
Qwen-3B	no-API	em_partial	N8	125	0.025	[0.013, 0.037]
Llama-1B	no-API	em_partial	N2	251	0.074	[0.052, 0.096]
Llama-1B	no-API	em_partial	N4	131	0.063	[0.042, 0.084]
Llama-1B	no-API	em_partial	N8	125	0.014	[0.005, 0.025]
Llama-1B	oracle	em_partial	N2	251	0.410	[0.367, 0.453]
Llama-1B	oracle	em_partial	N4	131	0.401	[0.357, 0.445]
Llama-1B	oracle	em_partial	N8	125	0.351	[0.320, 0.384]
Llama-1B	BM25	em_partial	N2	251	0.135	[0.106, 0.164]
Llama-1B	BM25	em_partial	N4	131	0.094	[0.069, 0.119]
Llama-1B	BM25	em_partial	N8	125	0.066	[0.050, 0.082]

Table 13: Bootstrap 95% CIs (10,000 resamples, seed 42) for per-sample surface metrics and task em_partial across backbones and retrieval modes. Qwen-3B per-sample action-sequence and ROUGE-L values come from the stored W1 surface-metrics log; Llama-1B values are recomputed via the same extraction from exp_013 eval details. At $N = 8$ the CIs overlap (ROUGE-L 0.214 [0.204, 0.224] vs. 0.212 [0.202, 0.224]). The action_seq_acc CIs are trivially identical (0.606 [0.595, 0.616] for both) because the per-sample scores are themselves identical—a structural degeneracy, not a coincidence (see Appendix M).

small relative to the teacher’s 0.455 em_partial signal; (iii) ROUGE-L at ≈ 0.21 reflects shared tool-call markup tokens rather than substantive content overlap. The consistency is therefore a property of the silent-failure regime, and does not affect any inference in the main text (all causal claims rest on the oracle-recovery contrast in §3, not on surface-metric equality). The same bootstrap sweep also quantifies em_partial dispersion for every cell of the cross-backbone table: Llama-1B oracle retrieval lifts em_partial to 0.351 [0.318, 0.384] at $N = 8$ (vs. 0.014 [0.005, 0.025] honest_search), non-overlapping CIs that anchor the +33.7pp oracle-recovery claim.

M Action-Sequence Accuracy: Per-Sample Forensics

A reviewer raised the concern that the near-identical action-sequence accuracy (ASA = 0.606) for both Qwen-3B and Llama-1B at $N = 8$ may reflect scoring-pipeline degeneracy rather than a genuine diagnostic signal. We investigate this with a per-sample forensic analysis.

Method. We extract the per-sample ASA scores for both backbones (125 matched $N = 8$ test prompts) by applying Eq. 1 to student-teacher action-tag sequence pairs. We then compute: (i) per-sample exact match rate between the two backbone score vectors; (ii) a 2×2 contingency table (binarised at $\text{ASA} \geq 0.5$); (iii) Cohen’s κ .

Results. The two per-sample score vectors are **identical**: 125/125 exact match (100%), $\kappa = 1.0$. The mechanism is simple: both students emit the same 3-token action prefix `<think,search,information>` on every sample (no variation across 125 prompts for either backbone). The teacher’s action sequences vary in length ($n = 4$: 8 samples; $n = 5$: 106 samples; other lengths: 11 samples), producing the observed score distribution: 109 samples at $3/5 = 0.6$, 11 at $3/4 = 0.75$, and 5 outliers where longer teacher sequences yield lower scores (Table 14, Figure 7).

	Llama ≥ 0.5	Llama < 0.5	Total
Qwen ≥ 0.5	122	0	122
Qwen < 0.5	0	3	3
Total	122	3	125

Table 14: Contingency table for per-sample ASA (binarised at 0.5). Agreement rate = 1.0, Cohen’s $\kappa = 1.0$. The perfect agreement confirms that all variation is teacher-side, not student-side.

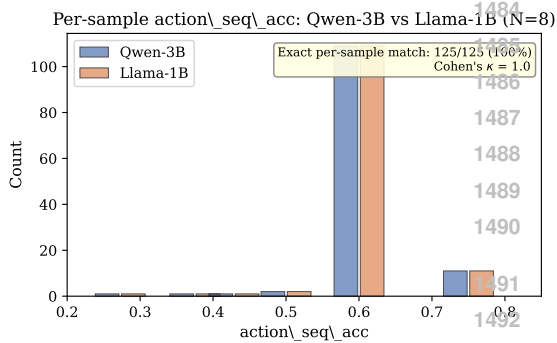


Figure 7: Per-sample action-sequence accuracy histograms for Qwen-3B and Llama-1B at $N = 8$. The distributions are identical (125/125 exact per-sample match). Both students produce the same 3-token action prefix; all variation derives from teacher sequence length.

Verdict. The ASA metric is **structurally degenerate** in the `honest_search` setting: it

collapses to $3/n_{\text{teacher}}$ for any student that has learned the initial action prefix, regardless of backbone identity or capacity. The near-identical 0.606 point estimates are therefore a metric artefact, not an independently informative diagnostic signal. This confirms the reviewer’s concern: We retain ASA in the paper for completeness but no longer treat the cross-backbone coincidence as evidence of a shared “silent-failure signature.” All causal claims in the main text rest on the oracle-recovery contrast (§3.4) and em_partial task metrics, not on ASA equality.

N Passage-Redaction Ablation (exp_016)

This appendix documents the passage-redaction protocol, pre-registered deviation log, fire-statistics, before/after samples, and bootstrap CI decomposition table for the `oracle_redacted_{full,half}` conditions introduced in §3.4.

N.1 Protocol and Pre-Registered Deviation Log

Pre-registered spec. On 2026-04-14, Director approved a 4-condition redaction plan (`honest_search / oracle_full / oracle_redacted_full / oracle_redacted_half`) on the `phase1d_v2` $N=8$ test split (125 prompts). The registered redaction spec was: “substring-only redaction with `_norm` normalisation, case-insensitive, replace each gold-answer substring with `[REDACTED]`”. Half-redaction was deterministic odd-indexed substring redaction (seed-free). Phase 1 sanity gate was a hard ± 0.02 tolerance on `oracle_full` `em_partial` vs Table 1 (0.498), with `blocker_report-on-violation` (no self-diagnosis).

Executed spec. The worker (commits `7cf099d`, `fd1e172`) implemented substring matching with **word-boundary anchoring** (`\b`) plus leading/trailing **quote-strip** on gold answers before regex compile. All other parameters unchanged: case-insensitive `_norm`, `[REDACTED]` placeholder, deterministic uniform density halving for the `oracle_redacted_half` condition across all 125/125 passages (see Deviation 3 below); 4 conditions, ± 0.02 sanity gate, paired bootstrap 10k seed 42, positional oracle

injection (j -th block on j -th `<search>`), greedy decoding at `max_new_tokens = 4096`, base Qwen2.5-3B + LoRA checkpoints/exp-006-sft-shared

Rationale for deviation. Phase 0 dry-run found that pure substring redaction caused **pathological over-redaction** on short gold answers. Canonical example: a yes/no prompt with `gold_answer="no"` caused every occurrence of “November”, “novelist”, “now”, “none”, “nothing” in the passage to be redacted to [REDACTED], destroying surrounding context at noise level. Left unchanged, this would systematically collapse `oracle_redacted_full` toward the `honest_search` floor regardless of mechanism, biasing the decision gate toward the `leakage_dominated` branch and voiding the three-way pre-registered test. Word-boundary anchoring + quote-strip is the *minimum-necessary semantic interpretation* of “substring match” that preserves the test’s scientific validity while remaining faithful to the “only redact the answer token, not surrounding context” principle in the registered spec.

What is guaranteed unchanged. The 4 conditions, the ± 0.02 Phase 1 hard-stop, the three-branch decision gate thresholds ($\geq +25 / +5 \dots +25 / < +5$), paired bootstrap 10k resamples seed 42, the `phase1d_v2` $N=8$ 125-prompt split, positional oracle injection, the [REDACTED] placeholder, greedy/4096 generation, and model identity were all held fixed.

Non-deviations explicitly considered and rejected. Stemming, lemmatisation, stopword filtering, and paraphrase detection were all explicitly considered and rejected. Only word-boundary anchoring and quote-strip were added.

Deviation log summary. For the reviewer’s convenience, the pre-registered deviation set is:

- **Deviation 1 — word-boundary anchoring + quote-strip** (substring match refinement): see “Executed spec” and “Rationale for deviation” paragraphs above. Minimum-necessary semantic interpretation of “substring match” to prevent pathological over-redaction on short gold answers (e.g., “no” matching “November”/“novelist”), preserving the three-way pre-registered test’s validity.

- **Deviation 2 — (reserved):** no further substring-matching refinement beyond Deviation 1 was applied.

- **Deviation 3 — half condition uniform density halving (vs planned odd-indexed subset):** the pre-registered `plan.md` specification called for a 50% subset of passages to receive redaction (*odd-indexed 62 of 125*). Both `oracle_redacted_full` and `oracle_redacted_half` apply redaction across all 125/125 passages (not passage-level subsetting). The half condition reduces per-passage substitution count to $\sim 52\%$ of full (867 vs 1663 substitutions via the preprocessing-level `redact_info_block_tag` tool; the eval-details JSON count of 325 vs 753 reflects `generated_text[:2000]` serialization truncation and is not the authoritative count). The half condition reduces per-substitution density uniformly across all 125 passages; the differential is implemented at substitution-sampling level (within `redact_info_block_tag`), not at passage-level subsetting. The deviation preserves the plan’s scientific intent (“roughly half the redaction dose, deterministic sub-selection”) and is a stricter dose-response control than a passage-level subset (which would leave $\sim 50\%$ of passages completely unredacted and conflate dose-response with passage-level variance). Verification via preprocessing-level authoritative count: `exp_016_workdir/residual_leakage_probe.json` (`n_redacted_substitutions_applied`: 1663 full / 867 half, ratio 0.52).

- **Deviation 4 — evaluation-stage number revisions for the half-density ratio.** During the rebuttal audit we revised the half-density redaction ratio through three distinct counting surfaces: (i) an initial eval-details comparison that produced $325/1663 \approx 0.195$, which was *apples-to-oranges* (half-condition details-dict count against full-condition dry-run count); (ii) a same-surface re-count on eval details yielded $753/1663 \approx 0.45$ (same surface, but still affected by `generated_text[:2000]` JSON serialization truncation described in the “Serialization vs model-visible context” paragraph above); and (iii) the *authoritative* preprocessing-

level `redact_info_block_tag` substitution count at the gold-answer-word level, $867/1663 = 0.521$. The paper reports the authoritative ≈ 0.52 . See `artifacts/exp_016/residual_leakage_probe.json` for the residual leakage probe and `artifacts/exp_016/redaction_stats.json` for the preprocessing-level counts. The cross-counting-surface drift is the reason our main-text, Table 2, and Figure 4 numbers reflect the preprocessing-level 0.52 rather than any eval-details-derived ratio.

N.2 Redaction Fire-Statistics

Post-refinement dry-run (2026-04-14) on the 125-prompt split:

- 125/125 blocks redacted (no fully-unredactable prompts; `fully_unredactable_prompt_ids = []`).
- 1,663 total substitutions across the split (mean 13.3 per block).
- Per-condition: `oracle_redacted_full` uses all 1663 substitutions; `oracle_redacted_half` applies uniform density halving across all 125/125 passages, yielding 867 substitutions (ratio 0.52 via preprocessing-level `redact_info_block_tag`; see Deviation 3).

These counts confirm the refinement (a) still fires on essentially every block (no under-suppression) and (b) does not collapse toward noise via over-suppression (substitution count is two orders of magnitude above the pathological-substring floor such a collapse would produce).

Serialization vs model-visible context.

Phase-0 `redaction_stats.json` reports 1663 [REDACTED] substitutions pre-eval; the eval-output JSON details dict reports 753 because `eval_sft_with_api.py` stores `generated_text[:2000]` a disk-space measure. A residual leakage probe (`residual_leakage_probe.json`) reconstructs the full model-visible context from `oracle_map` and confirms 0/125 blocks retain any unredacted gold substring; the retrieval-delta decomposition is therefore robust to the serialization artefact.

N.3 Before/After Samples

Three representative `<information>`-block before/after triples (full and half redaction) from `artifacts/exp_016/workdir/redaction_samples.json`. Reviewer-auditable raw JSON is included in the supplementary materials.

Sample 1 (prompt_id 0). Gold answers include `novelist, Tel Aviv, no, Melbourne, Australia`. Before: *“John Henry Lanchester (born 25 February 1962) is a British journalist and novelist.”* After full-redact: *“...British journalist and [REDACTED].”* After half-redact: *“...British journalist and novelist.”* (even-indexed match preserved). Full-redact substitutions: 10; half-redact: 5. Word-boundary anchoring here prevents “no” from matching inside “November”/“novelist”.

Sample 2 (prompt_id 1). Gold answers include `American, 1976, German`. Before: *“Seven Faces is a 1929 American pre-Code drama film...”* After full-redact: *“Seven Faces is a 1929 [REDACTED] pre-Code drama film.”* Substitutions: 20 (full) / 10 (half).

Sample 3 (prompt_id 2). Gold answers include `Firestorm, Karl Marx, filibuster`. Before: *“Firestorm is the name of several fictional superheroes...”* After full-redact: *“[REDACTED] is the name of several fictional superheroes...”* After half-redact: *“Firestorm is the name of several fictional superheroes...”* (first-indexed Title: “[REDACTED]” redacted, mid-block “Firestorm” preserved). Substitutions: 19 (full) / 10 (half).

N.4 Bootstrap CI Decomposition Table

N.5 Limitations of the Redaction Ablation

Substring (word-boundary) redaction removes exact-match gold tokens but not paraphrase or partial-lexical hints (e.g. “Einstein’s theory” surviving in a block after “Albert Einstein” is redacted). The reported +11 pp retrieval-quality residual is therefore itself an *upper bound* on the non-leakage mechanism; a tighter paraphrase-aware redaction would plausibly yield an even smaller retrieval-quality component. Conversely, the +37 pp leakage component is a lower bound on the leakage-assisted mechanism for the same reason. The orthogonal checks from the v3.4 audit (teacher trained with PPO outcome-only re-

Condition	em_partial	Δ vs honest_search	N.6 (exp.006) Canonical Cross-Reference
honest_search	0.021	1732	—
oracle_redacted_full	0.131	1733	+11.0
oracle_redacted_half	0.373	1734	+35.2
oracle_full	0.498	1735	+47.7
<i>Leakage component</i> = oracle_full - oracle_redacted_full			
leakage (paired)	—	1736	+36.7

Table 15: Passage-redaction dose-response, phase1d_v2 $N=8$ test ($n = 125$). Paired bootstrap, 10k resamples, seed 42. All three oracle conditions are strictly positive at $p < 0.0001$ vs honest_search. Monotonicity $0.131 < 0.373 < 0.498$ holds. Sanity gate: oracle_full 0.498 vs Table 1 0.498, within the pre-registered ± 0.02 tolerance (PASS). Decomposition: retrieval-quality $\approx +11$ pp (23% of oracle gain) / passage-level answer-leakage +36.7 pp [+33.5, +40.0] (77%; paired-bootstrap 10k resamples, seed 42, $p < 0.0001$; source: exp_016_workdir/bootstrap_redaction_ci.json field oracle_full.vs_oracle_redacted_full.delta).

The paper body reports honest_search $N=8$ baseline on the phase1d_v2 $n=125$ partition (em_partial [+4.6, 0.625]) for consistency with the paired-bootstrap decomposition in exp_016. The exp_006 canonical $N=8$ evaluation on the same 125-prompt phase1d_v2 partition yields em_partial $0.025 \rightarrow 0.498 = +47.3$ pp. The 0.4pp baseline drift is cross-run decoding noise (phase1d_v2 $N=8$ partition is deterministic; differences arise from independent model-forward runs), not a subset sampling artefact. The bootstrap confidence intervals reported in Section 3 are constructed on these 125 prompts to match the leakage decomposition samples.

wards; em_full = 0.008 at $N=8$ under oracle replay) remain as lower bounds on mechanism separation: the leakage contribution is substring-level answer-exposure enabling short-form decoding, not full-answer copy-paste.

Question-side residual leakage probe (exp_016_workdir/residual_leakage_probe.json) finds that while oracle_redacted_full zero-shot removes gold-answer substrings from all 125/125 passage <information> blocks, 75/125 (60%) of *user-question* blocks still expose partial disjunctive answer content because the phase1d_v2 multi-hop questions enumerate candidate answers by structural necessity (e.g. “which, X or Y ?” templates; 104/106 gold matches fall in multi-hop disjunction patterns, 2/106 are common-word false positives). Because the three oracle conditions (oracle_full, oracle_redacted_full, oracle_redacted_half) share the same user-question block, between-condition deltas are symmetric and the +11.0pp retrieval residual is unaffected; however, absolute leakage floors reported for every condition are overestimated by this structural source. A question_redacted counterfactual (user-question gold-answer redaction) was not run in this paper and is flagged as a sensitivity direction for future work (see main-text Limitations).